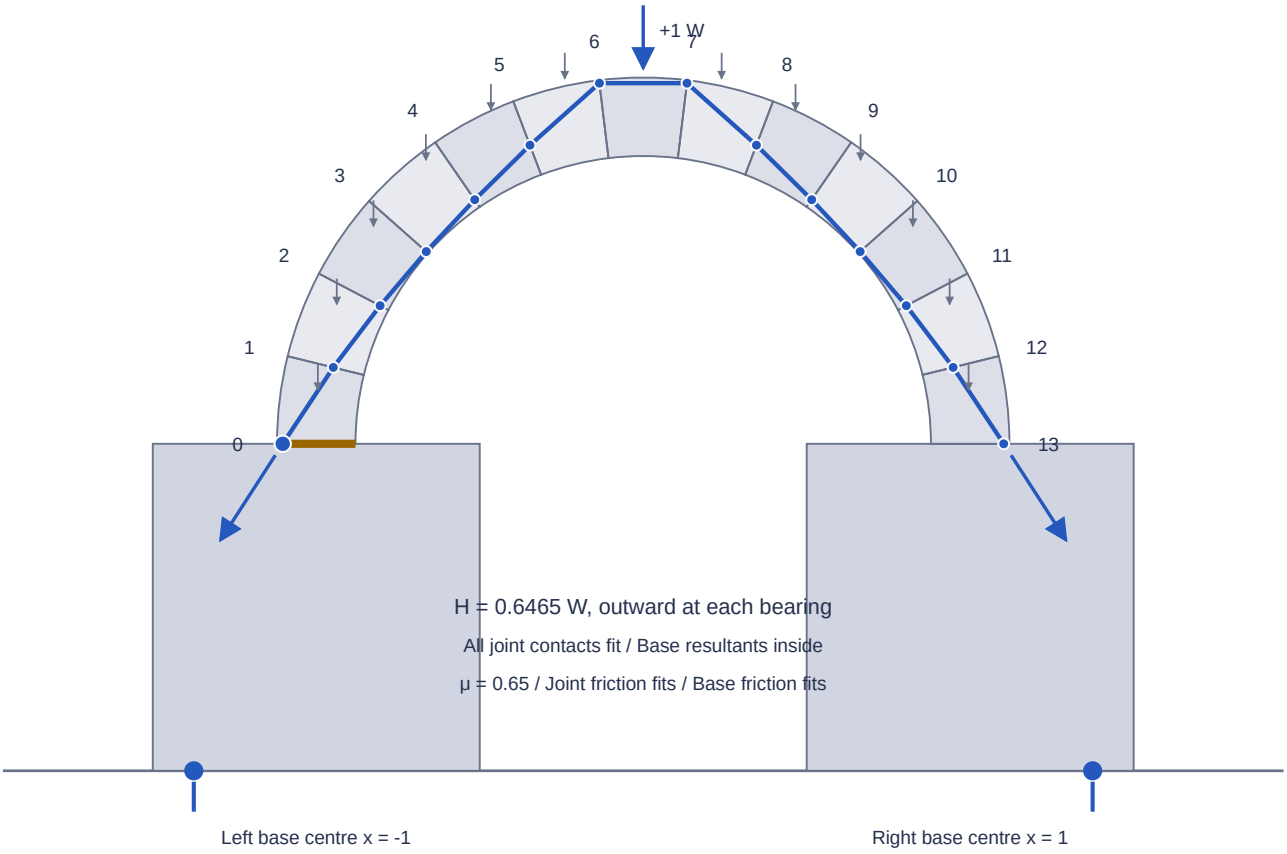
The arch and the pier

This path meets the contact and friction checks.

Modern 13-block symmetric equilibrium experiment; no tension, unlimited compression, static Coulomb friction |T| <= μN with uniform illustrative μ = 0.65 at every joint and both pier bases. Not a measured ancient material, an ancient design rule or a structural safety assessment.

Your experiment



Arch shape

Centreline span 2; rise 1; radial thickness 0.24. Length unit: one half-span.

Load

Own weight W = 1; extra 1 W on the crown stone. Every geometry is renormalized to W = 1.

Trial

Horizontal thrust 0.64646079 W; left joint contact 92.70259846% from the inner edge. C = 1.10248624 W half-span.

Each pier

Base width 1; height 1; weight 1 W. Base resultant 0.374474 outward from its centre; available half-width 0.5.

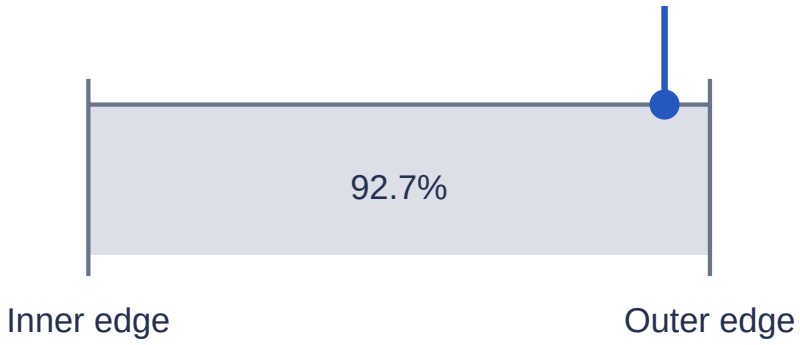
Sliding assumption

Uniform illustrative μ = 0.65. Joint friction passes; base friction passes. For this trial, the greatest required |T|/N is 0.64646079. This is not a minimum over all paths or a measured material value.

The blue line joins interface contacts to guide the eye. It is not a continuous stress solution inside the stones.

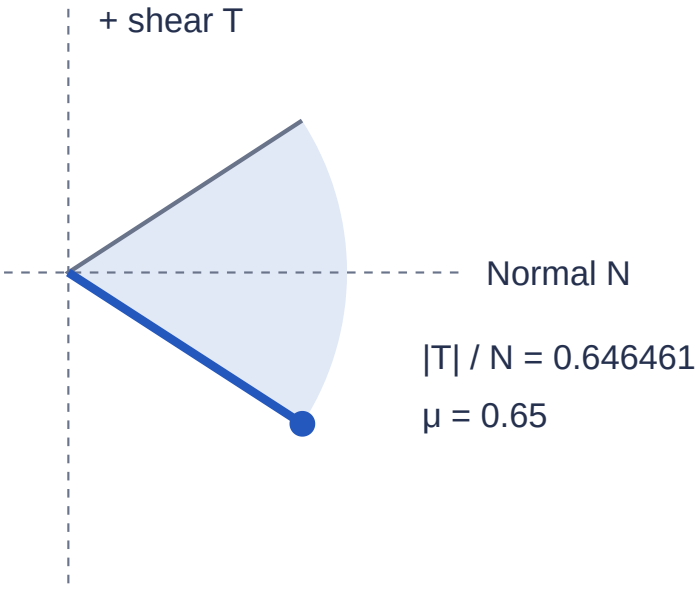
Contact need not stay within the middle third: joints may open partly under this ideal model. Edge contact is a limiting state with no finite stress solution. Finite static friction is checked at all fourteen joints and both bases with the chosen illustrative coefficient. Material strength is not evaluated.

Selected joint 0, rotated and enlarged



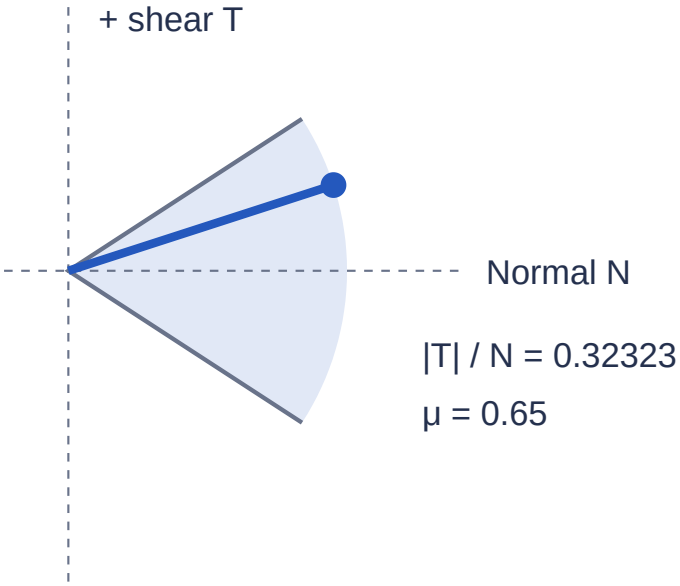
Force directions and sliding

Joint 0



Directions only; force sizes in W below.

Either pier base



Directions only; force sizes in W below.

Shaded cones show |T| ≤ μN. The ray direction, not its length, is compared with the cone. No cone is drawn when sliding is excluded by assumption. At a base, N = V + P and |T| = H. Increasing pier weight can help base friction; increasing width alone cannot. These are static equilibrium tests, not a simulation of motion after failure.

Joint	Contact from inner edge	Normal force, W	Shear, W	Contact	Required |T|/N	Friction
0	92.7026%	1	-0.646461	Inside	0.646461	Within limit
1	40.0813%	1.050962	-0.406769	Inside	0.387044	Within limit
2	11.9689%	1.049657	-0.179185	Inside	0.170708	Within limit

Joint	Contact from inner edge	Normal force, W	Shear, W	Contact	Required |T|/N	Friction
3	2.6837%	1.00446	0.026212	Inside	0.026095	Within limit
4	11.257%	0.925302	0.202526	Inside	0.218876	Within limit
5	40.2568%	0.82267	0.346156	Inside	0.420772	Within limit
6	96.1854%	0.706652	0.456613	Inside	0.646165	Within limit
7	96.1854%	0.706652	-0.456613	Inside	0.646165	Within limit
8	40.2568%	0.82267	-0.346156	Inside	0.420772	Within limit
9	11.257%	0.925302	-0.202526	Inside	0.218876	Within limit
10	2.6837%	1.00446	-0.026212	Inside	0.026095	Within limit
11	11.9689%	1.049657	0.179185	Inside	0.170708	Within limit
12	40.0813%	1.050962	0.406769	Inside	0.387044	Within limit
13	92.7026%	1	0.646461	Inside	0.646461	Within limit

Read Vitruvius

Latin: F. Krohn's 1912 edition, printed p. 141. New English and notes for Gonia. Joined line-end hyphens; sentence punctuation adapted for parallel reading. Readings 3 and 4 form one sentence. These passages describe relieving arches and their supports, not the equations of modern limit analysis.

An arch can relieve a wall's weight

item administrandum est, uti levent onus parietum fornicationes cuneorum divisionibus et ad centrum respondentes earum conclusurae.

Care must also be taken that arches, with their divisions into wedges and their closures directed toward the centre, relieve the weight of the walls.

uti levent states the intended result: ‘that they relieve’. onus is the object, ‘weight’; fornicationes is the subject, ‘arches’. ad centrum refers to radial construction, not to the modern blue force path. Krohn's apparatus records variant readings at cuneorum.

The timber underneath can be replaced

cum enim extra trabes aut liminum capita arcus cuneis erunt conclusae, primum non pandabit materies levata onere; deinde, si quod vetustate vitium ceperit, sine molitione fulturarum faciliter mutabitur.

For when arches have been closed with wedges above the beams or the ends of the lintels, first the timber, relieved of its load, will not sag; then, if it develops a defect through age, it will be easily replaced without constructing props.

levata onere is ‘relieved of the load’. The passage describes a relieving arch over timber. Our isolated arch omits that timber and the wall above it; it is not a reconstruction of this arrangement. Krohn prints the unusual arcus … conclusae; the reading is retained.

The outer piers need more breadth

itemque, quae pilatim aguntur aedificia et cuneorum divisionibus coagmentis ad centrum respondentibus fornices concluduntur, extremae pilae in his latiores spatio sunt faciundae, uti vires eae habentes resistere possint,

Likewise, in buildings carried on piers, where arches are closed with wedge divisions and joints directed toward the centre, the outer piers must be made broader in extent, so that, having strength, they can resist,

The sentence continues in the next reading. latiores is comparative, ‘broader’; sunt faciundae expresses an obligation, ‘must be made’. There is no numerical pier-width formula here. Our width and mass controls are modern independent choices.

The loaded wedges can push the bearings out

cum cunei ab oneribus parietum pressi per coagmenta ad centrum se prementes extruderent incumbas.

since the wedges, pressed by the loads of the walls and pressing against one another through the joints toward the centre, would push out the bearings.

This completes the preceding sentence. pressi is passive, ‘pressed’; se prementes is active and reciprocal here, ‘pressing against one another’. extruderent is ‘would push out’. The Latin describes an effect; it does not supply our equilibrium equations or assumptions.

Restraint belongs to the whole work

itaque si angulares pilae erunt spatiosis magnitudinibus, continendo cuneos firmitatem operibus praestabunt.

And so, if the corner piers are of ample size, by restraining the wedges they will provide stability to the works.

continendo is an ablative gerund, ‘by restraining’; cuneos is what is restrained. firmitatem is the object of praestabunt, ‘they will provide stability’. It is a claim about support, not a certificate for a particular surviving arch.

Words in this passage

cuneus · cunei · cuneos
wedge · wedges · wedges as object. A wedge-shaped stone in this context. cunei is nominative plural in the pushing clause; cuneos is accusative plural after continendo in the final clause.

coagmentum · coagmenta
joint · joints. The meeting of pieces. per coagmenta means ‘through the joints’. The model checks where a resultant crosses each radial joint; it tests joint friction only in the optional modern finite-friction model.

centrum · ad centrum
centre · toward the centre. The common centre of the radial divisions. In a shallow circular arch this centre lies below the spring level. It is a geometric centre, not a centre of gravity.

pila · pilae
pier · piers. A massive support here, rather than the word for a slender column, columna. extremae pilae are the outer piers; angulares pilae are the corner piers.

onus · onera · oneribus
weight or load · loads · by loads. One word changes form with its job in the sentence. ab oneribus parietum is ‘by the loads of the walls’. We use W for the arch's own weight and label any extra load separately.

latiores
broader. Comparative adjective agreeing with pilae. The author asks for more breadth, but the passage gives no particular ratio. Base width and weight are kept separate on the bench.

incumbas
bearings or supporting parts. Accusative plural, the object of extruderent: the parts that would be thrust out. ‘Abutments’ is a useful architectural rendering, not a claim that the word specifies our support shape.

firmitatem
stability or strength. Accusative singular of firmitas. The author says the piers provide it by restraining the wedges. Our much narrower result is equilibrium under explicit ideal assumptions.

Assumptions and a reproducible model

Thirteen equal-angle annular sectors. Self-weight acts at each exact area centroid and sums to W = 1. Symmetric vertical extra loads have their own stated magnitudes. Symmetry fixes the vertical reaction at each spring to half the total load. At each radial joint the resultant is located from force and moment equilibrium. A complete interval calculation tests every inner/outer edge pair and, in finite-friction mode, intersects the two Coulomb shear bounds at each joint and base. It does not sample trial paths.

Every pier has a symmetric base, width b, height h = 1 and independent weight P. Its base resultant's outward eccentricity is e = (C + Hh - V)/(V + P). It fits when |e| is at most b/2. The support head is shaped to meet the spring joint; its weight is idealized on its centreline.

Modern 13-block symmetric equilibrium experiment; no tension, unlimited compression, static Coulomb friction |T| <= μN with uniform illustrative μ = 0.65 at every joint and both pier bases. Not a measured ancient material, an ancient design rule or a structural safety assessment. There is no mortar, soil bearing, settlement, crushing, elastic response, earthquake, construction sequence or three-dimensional restraint. The finite interface check is not a continuous stress solution. No dimensions or loads are a survey of a historical structure. A failed trial does not prove that every path fails. A fitted path is one equilibrium witness under these assumptions.

Full model and formulas. Heyman, The stone skeleton (1966); ETH Zurich, Arch bridges (2025 lectures). Drake: static Coulomb friction. Modern illustrations are not reproduced.

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culum inferiorum medio conlocentur, uti solido respon-
deant; nam si in pendentibus onera fuerint parietum aut
columnarum, non poterint habere perpetuam firmitatem.
praeterea inter lumina secundum pilas et antas postes si 2
supponentur, erunt non vitiosae. limina enim et trabes
structuris cum sint oneratae, medio spatio pandantes fran-
gunt sub lysi structuras; cum autem subiecti fuerint et
subcuneati postes, non patiuntur insidere trabes neque
eas laedere. item administrandum est, uti levent onus 3
parietum fornicationes cuneorum divisionibus et ad cen-
trum respondentes earum conclusurae. cum enim extra
trabes aut liminum capita arcus cuneis erunt conclusae,
primum non pandabit materies levata onere; deinde, si
quod vetustate vitium ceperit, sine molitione fulturarum
faciliter mutabitur. itemque, quae pilatim aguntur aedificia 4
et cuneorum divisionibus coagmentis ad centrum respon-
dentibus fornices concluduntur, extremae pilae in his la-
tiores spatio sunt faciundae, uti vires eae habentes re-
sistere possint, cum cunei ab oneribus parietum pressi per
coagmenta ad centrum se prementes extruderent incumbas.
itaque si angulares pilae erunt spatiosis magnitudinibus,
continendo cuneos firmitatem operibus praestabunt.

Cum in his rebus animadversum fuerit, uti ea diligentia 5
153 in his adhibeatur, non minus etiam observandum est, uti
omnes structurae perpendiculo respondeant neque habeant
in ulla parte proclinationes. maxima autem esse debet
cura substructionum, quod in his infinita vitia solet facere
terrae congestio. ea enim non potest esse semper uno
pondere, quo solet esse per aestatem, sed hibernis tem-
poribus recipiendo ex imbribus aquae multitudinem cre-
scens et pondere et amplitudine disrumpit et extrudit struc-
turarum saeptiones. itaque, ut huic vitio medeatur, sic 6
erit faciundum, ut primum pro amplitudine congestionis

---

3 poterint *HS*: poterunt *G* 8 insidera *H* 9 utile ventonus *HSG*^{c}, uti. Liventonus *G* 10 cuneorum *H*: cum eorum *GS* 15 facile *S* 25 perpendicula *G*

Vitruvius VI.8.1-6, Krohn 1912, p. 141. Radial wedges, relieving arches and wider outer piers. The complete apparatus remains below the text. PDF page 159. SHA-256 63c035dae4114c03c8404171f624f93690fe122eedd301e9913c0525cd7f781d.

autem hypaethrus ambulationes xysta appellant, quas 151
Graeci παραδρομιδας dicunt. item προθυρα graece dicuntur, quae sunt ante ianuas vestibula, nos autem appel-
6 lamus prothyra, quae graece dicuntur διαθυρα. item si
qua virili figura signa mutulos aut coronas sustinent, 5
nostri telamones appellant, cuius rationes, quid ita aut quare dicantur, ex historiis non inveniuntur, Graeci vero eos ατλαντας vocitant. Atlas enim formatur historia sustinens mundum, ideo quod is primum cursum solis et lunae siderumque omnium versationum rationes vigore animi 10
sollertiaque curavit hominibus tradenda, eaque re a pictoribus et statuariis deformatur pro eo beneficio sustinens mundum, filiaeque eius Atlantides, quas nos vergilias, Graeci autem πλειαδας nominant, cum sideribus in mundo
7 sunt dedicatae. nec tamen ego, ut mutetur consuetudo 15
nominationum aut sermonis, ideo haec proposui, sed uti (15)
non sint ignota philologis, exponenda iudicavi.

Quibus consuetudinibus aedificia italico more et Graecorum institutis conformantur, exposui et de symmetriis singulorum generum proportiones perscripsi. ergo quoniam 20
de venustate decoreque ante est conscriptum, nunc ex- (20)
ponemus de firmitate, quemadmodum ea sine vitiis permanentia ad vetustatem conlocentur.

VIII Aedificia quae plano pede instituuntur, si fundamenta eorum facta fuerint ita, ut in prioribus libris de muro et 25
theatris a nobis est expositum, ad vetustatem ea erunt (25)
sine dubitatione firma. sin autem hypogea concamarationesque instituentur, fundationes eorum fieri debent cras- 152
siores, quam quae in superioribus aedificiis structurae sunt futurae. eorumque parietes, pilae, columnae ad perpendi- 30

---

1 xysta *ed*: xysti *x* 2 paradromidas *x* prothyra *HS*, prothira *G* 3 ante ianuas *S*: ante inianuas *HG* (*cum* in *add.*) 4 graeci *H* diathyra *x* 6 cuius *HG* (*et in marg.* S^c): quorū *S* ratione *H* 8 atlantas *GS*, &lantas *H* vocitant *HG*: appellant *S* 9 his *H* 11 re *Joc*: res *x* 14 pliadas *x* 16 sermones *HS* uti *Ro*: ut ei *HG* (ut ea G^c), ut *S* 20 perscripsi. Ergo *GS*: perscribsi ero *H* 25 ut *GS*: aut *H* 26 a *G*: *om. HS* ea G^c: eae *x* 27 ypogea *x*

Vitruvius VI.7-8.1, Krohn 1912, p. 140. The chapter's opening and preceding context. Chapter VI.8 begins near the foot of this page. PDF page 158. SHA-256 8b96d1e71530256a99c36b6c1784678fcc8d8d4b484860aa6d61c60e23b69a44.

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Editable experiment


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